WhatsApp chat mood analyser

Course: Artificial Intelligence

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Our Team

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The problem

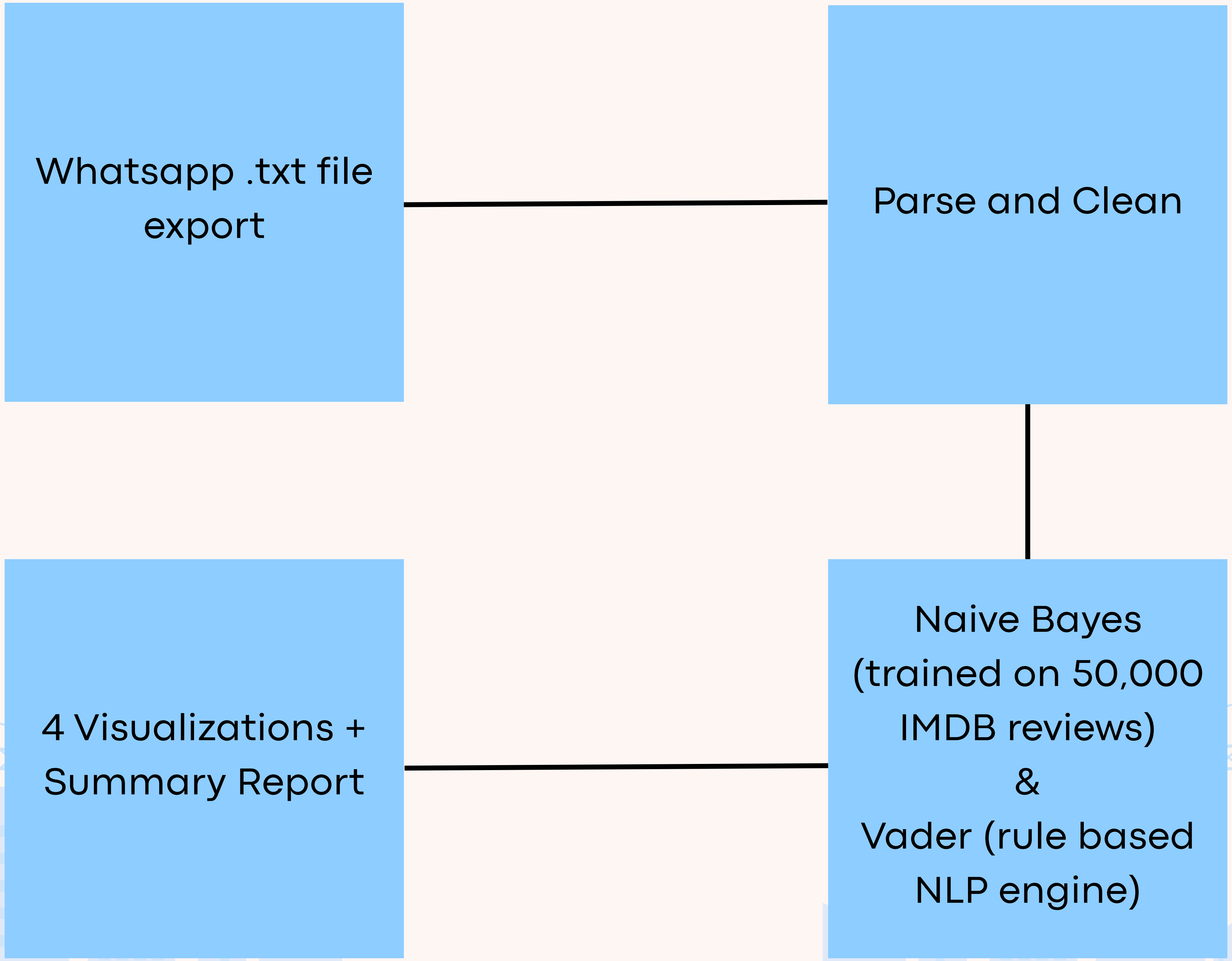
- WhatsApp processes 100 billion messages daily
- Manually reading chats to identify mood patterns is impractical
- No simple tool exists to analyze emotional tone of conversations

We have built a tool to solve this issue.

Objectives

- Parse real WhatsApp exported chat files automatically
- Train an ML model to classify message sentiment
- Apply a second NLP model for informal text
- Compare both models for stronger results
- Visualize mood patterns through charts

How it works


Whatsapp .txt file export
Parse and Clean
Naive Bayes (trained on 50,000 IMDB reviews) & Vader (rule based NLP engine)
4 Visualizations + Summary Report

IMDB Dataset- Exploratory Data Analysis (EDA)

Total reviews:	**50, 000**
Positive:	25,000 (50%)
Negative	25,000 (50%)
Missing values:	None
Duplicate values:	Negligible
Avg length (per review)	Around 230 words

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============================================================
EXPLORATORY DATA ANALYSIS - IMDB DATASET
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1. DATASET SHAPE:
   Rows: 50000, Columns: 2

2. MISSING VALUES:
review       0
sentiment    0
dtype: int64

3. DUPLICATE ROWS: 418

4. SENTIMENT DISTRIBUTION:
   positive: 25000 (50.0%)
   negative: 25000 (50.0%)

5. REVIEW LENGTH (word count):
   Shortest review : 4 words
   Longest review  : 2470 words
   Average length  : 231 words

6. AVERAGE LENGTH BY SENTIMENT:
   negative: 229 words on average
   positive: 233 words on average
```

Output screenshot of the EDA cell

Text pre-processing

4 Cleaning Steps:

1. Lowercase "Amazing" → "amazing"
2. Remove HTML "
" → removed
3. Remove Punctuation "loved it!!!" → "loved it"
4. Remove Stopwords "it was the best" → "best"

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ORIGINAL:
One of the other reviewers has mentioned that after watching just 1 Oz episode you'll be hooked. They are right, as this is exactly what happe
ned with me.<br /><br />The first thing that struck me abo

CLEANED:
one reviewers mentioned watching oz episode youll hooked right exactly happened methe first thing struck oz brutality unflinching scenes viole
nce set right word go trust show faint hearted timid show
```

Example of Original vs Cleaned text

Model 1: Naive Bayes

- Trained on 40,000 IMDB reviews (80% split)
- TF-IDF Vectorization — top 5,000 features
- Tested on 10,000 unseen reviews (20% split)
- Accuracy: 85.06%

Why Naive Bayes?

→ Fast, lightweight, proven on text classification

→ Gives us a quantifiable accuracy benchmark

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Training on: 40000 reviews
Testing on:  10000 reviews

Model Accuracy: 85.06%
```

Model 2: VADER

- Rule-based NLP — no training required
- Pre-built dictionary of 7,500+ sentiment-scored words
- Handles informal text features:
- Capitalization: "AMAZING" scores higher than "amazing"
- Punctuation: "great!!!" scores higher than "great"
- Negations: "not good" → correctly negative

Why VADER?
→ Designed for social media & informal text
→ Handles Roman Urdu mixed messages better

Model 2: VADER

	sender	message	vader_sentiment
0	Steve	yaar aaj ka din bohot acha tha	neutral
1	Sophia	haan bhai mujhe bhi bohot acha laga	neutral
2	Harry	exam mein fail ho gaya yaar bohot bura laga	negative
3	Steve	uff yaar that's terrible so sorry to hear that	negative
4	Sophia	don't worry Harry you will do better next time	positive
5	Harry	thanks yaar appreciate it feeling a bit better...	positive
6	Steve	movie dekhi kal? bohot amazing thi	positive
7	Sophia	haan yaar loved it absolutely brilliant	positive
8	Harry	boring thi mujhe toh bilkul pasand nahi aayi	negative
9	Steve	what! it was the best film i have seen in years	positive
10	Sophia	i agree with Steve it was fantastic	positive
11	Harry	okay okay maybe i was in a bad mood	negative
12	Steve	good morning everyone hope you all are well	positive
13	Sophia	good morning feeling great today	positive

Dual Model approach

Naive Bayes	VADER
Formal English	Informal/Slang text
Quantifiable accuracy	No training needed
Struggles with Roman Urdu	Fixed dictionary - can't learn new words

Using both models together improves accuracy and enables more comprehensive sentiment analysis.

Whatsapp parser

Input: Exported .txt file Format:

Sender message format:
DD/MM/YYYY, HH:MM

We use Python Regex to extract:
→ Date
→ Time
→ Sender
→ Message Text

Output: Structured Pandas DataFrame

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Total messages parsed: 33
         date   time  sender  \
0  12/03/2025  10:00   Steve
1  12/03/2025  10:01  Sophia
2  12/03/2025  10:02   Harry
3  12/03/2025  10:03   Steve
4  12/03/2025  10:04  Sophia
5  12/03/2025  10:05   Harry
6  12/03/2025  11:00   Steve
7  12/03/2025  11:01  Sophia
8  12/03/2025  11:02   Harry
9  12/03/2025  11:03   Steve

                                        message            datetime
0                 yaar aaj ka din bohot acha tha 2025-03-12 10:00:00
1            haan bhai mujhe bhi bohot acha laga 2025-03-12 10:01:00
2      exam mein fail ho gaya yaar bohot bura laga 2025-03-12 10:02:00
3     uff yaar that's terrible so sorry to hear that 2025-03-12 10:03:00
4     don't worry Harry you will do better next time 2025-03-12 10:04:00
5   thanks yaar appreciate it feeling a bit better... 2025-03-12 10:05:00
6              movie dekhi kal? bohot amazing thi 2025-03-12 11:00:00
7         haan yaar loved it absolutely brilliant 2025-03-12 11:01:00
8      boring thi mujhe toh bilkul pasand nahi aayi 2025-03-12 11:02:00
9     what! it was the best film i have seen in years 2025-03-12 11:03:00
```

WhatsApp Chat- Exploratory Data Analysis (EDA)

Performed on parsed chat data before running sentiment models.
Key checks conducted:
- Total message count & date range verified
- Participant activity distribution analyzed
- Average message length calculated
- Missing values confirmed: None

Conclusion: Data was clean and ready for sentiment analysis.

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   WHATSAPP CHAT — EXPLORATORY DATA ANALYSIS
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Total Messages    : 33
Date Range        : 12/03/2025 to 14/03/2025
Participants      : Steve, Sophia, Harry

Messages Per Person:
   Steve: 11 (33.3%)
   Sophia: 11 (33.3%)
   Harry: 11 (33.3%)

Avg Message Length : 7.6 words
Shortest / Longest : 5 / 12 words

Most Active Day   : 12/03/2025 (12 messages)

Missing Values    : 0 (none)
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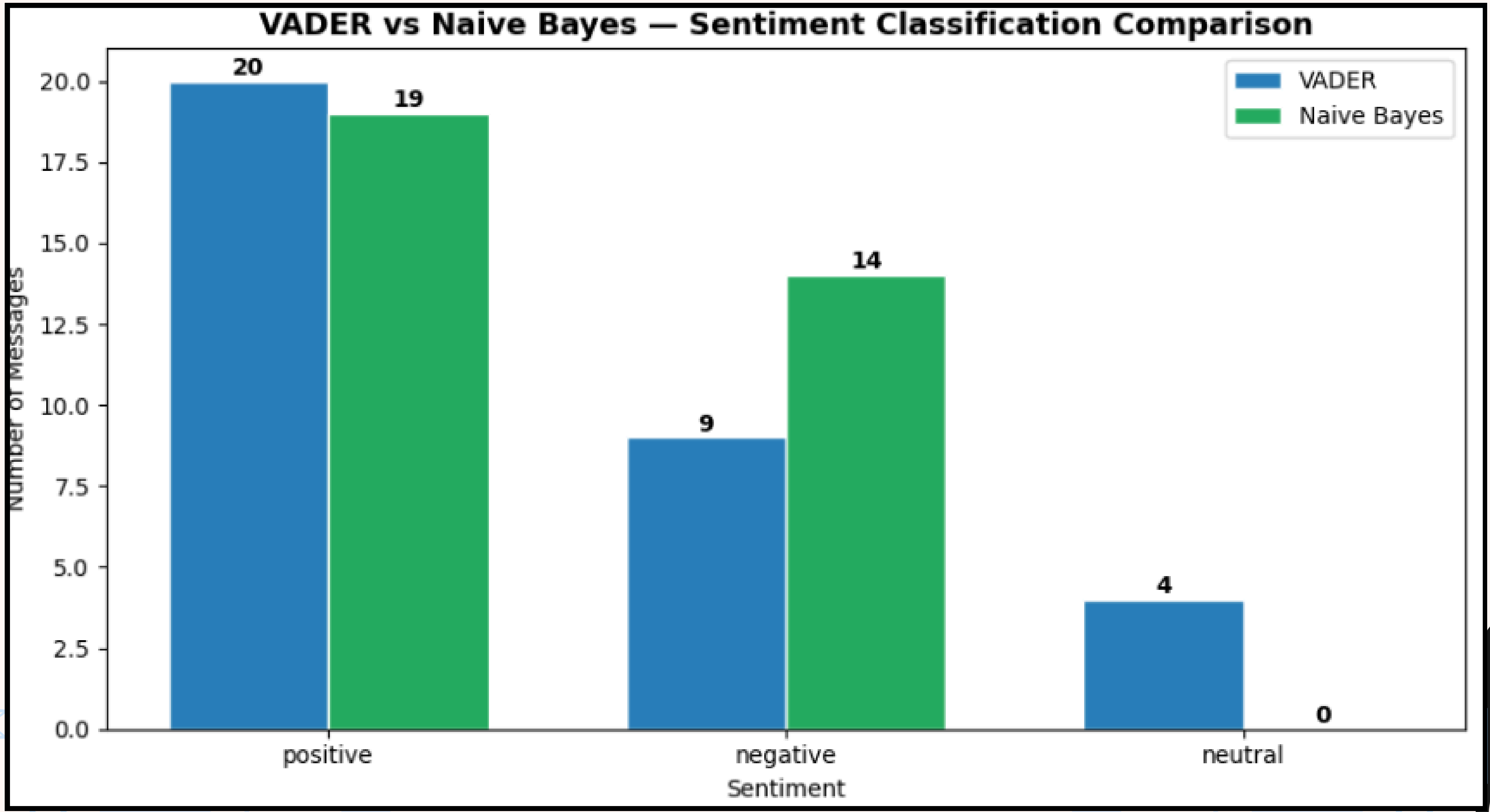
Model Comparison Chart



Note: Naive Bayes does not output 'neutral' — only positive/negative.

Results:

Chat 1:

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   WHATSAPP CHAT MOOD ANALYSIS REPORT
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Total Messages Analyzed: 33
Positive Messages: 20 (60.6%)
Negative Messages: 9 (27.3%)
Neutral Messages:  4 (12.1%)

Most Positive Person: Sophia
Most Negative Person: Harry

Naive Bayes Model Accuracy: 85.06%
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Chat 2:

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 'Parhai' WHATSAPP group CHAT MOOD ANALYSIS REPORT
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Total Messages Analyzed: 34
Positive Messages: 11 (32.4%)
Negative Messages: 7 (20.6%)
Neutral Messages:  16 (47.1%)

Most Positive Person: Bilal
Most Negative Person: Fawad

Naive Bayes Model Accuracy: 85.06%
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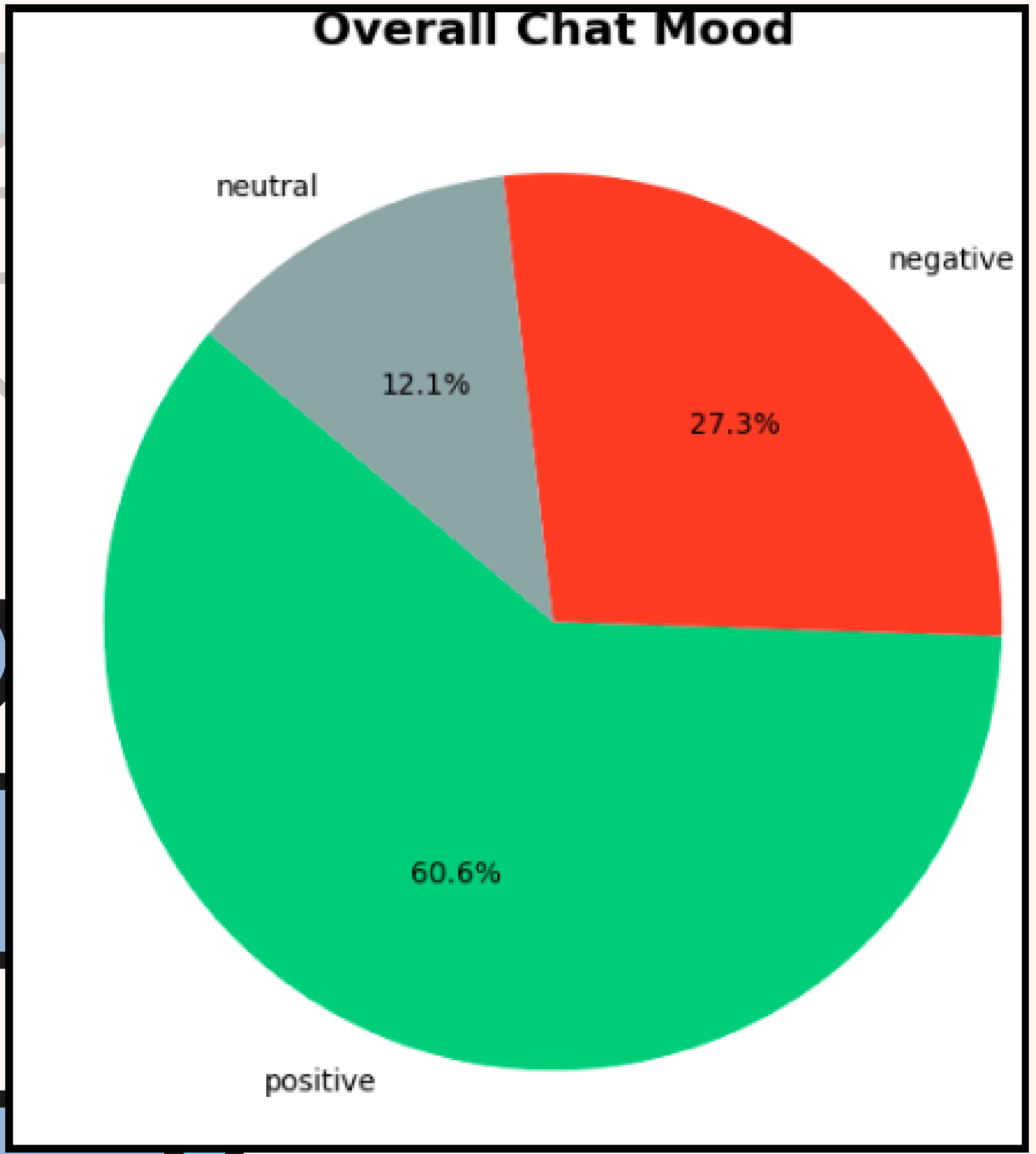
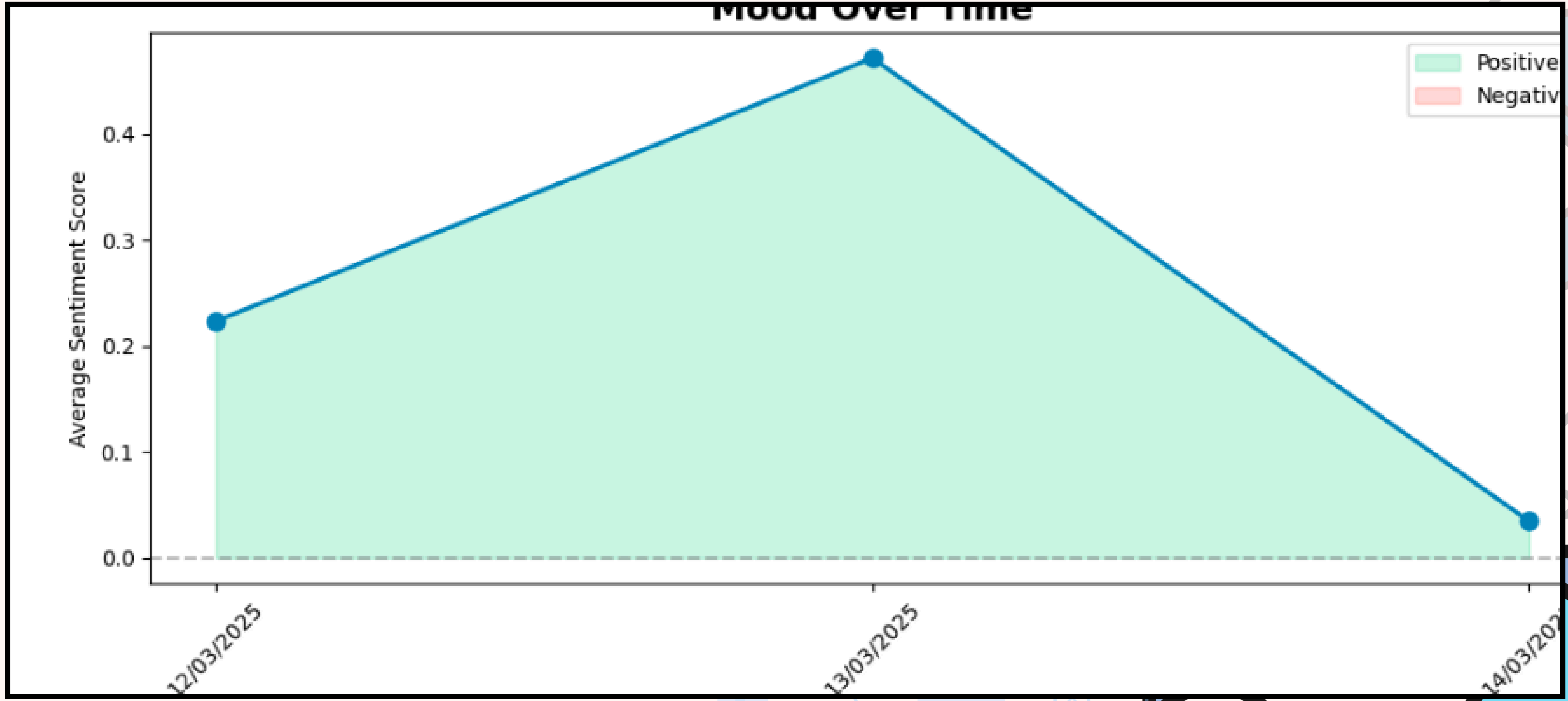
Visualisation 1 & 2 (chat 1)

Pie chart (overall mood)

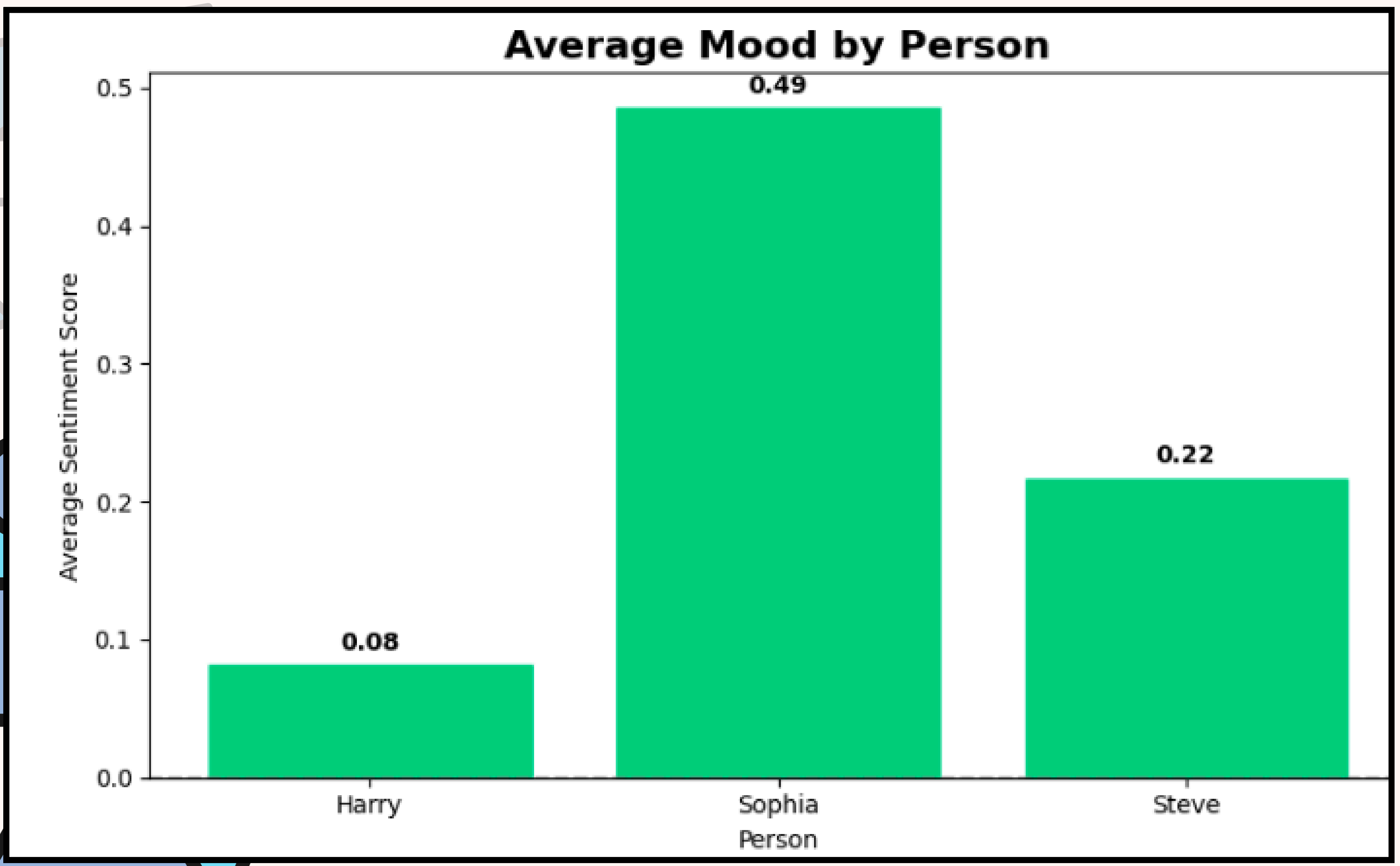


Line graph (mood over time)



Visualisation 3 & 4 (chat 1)

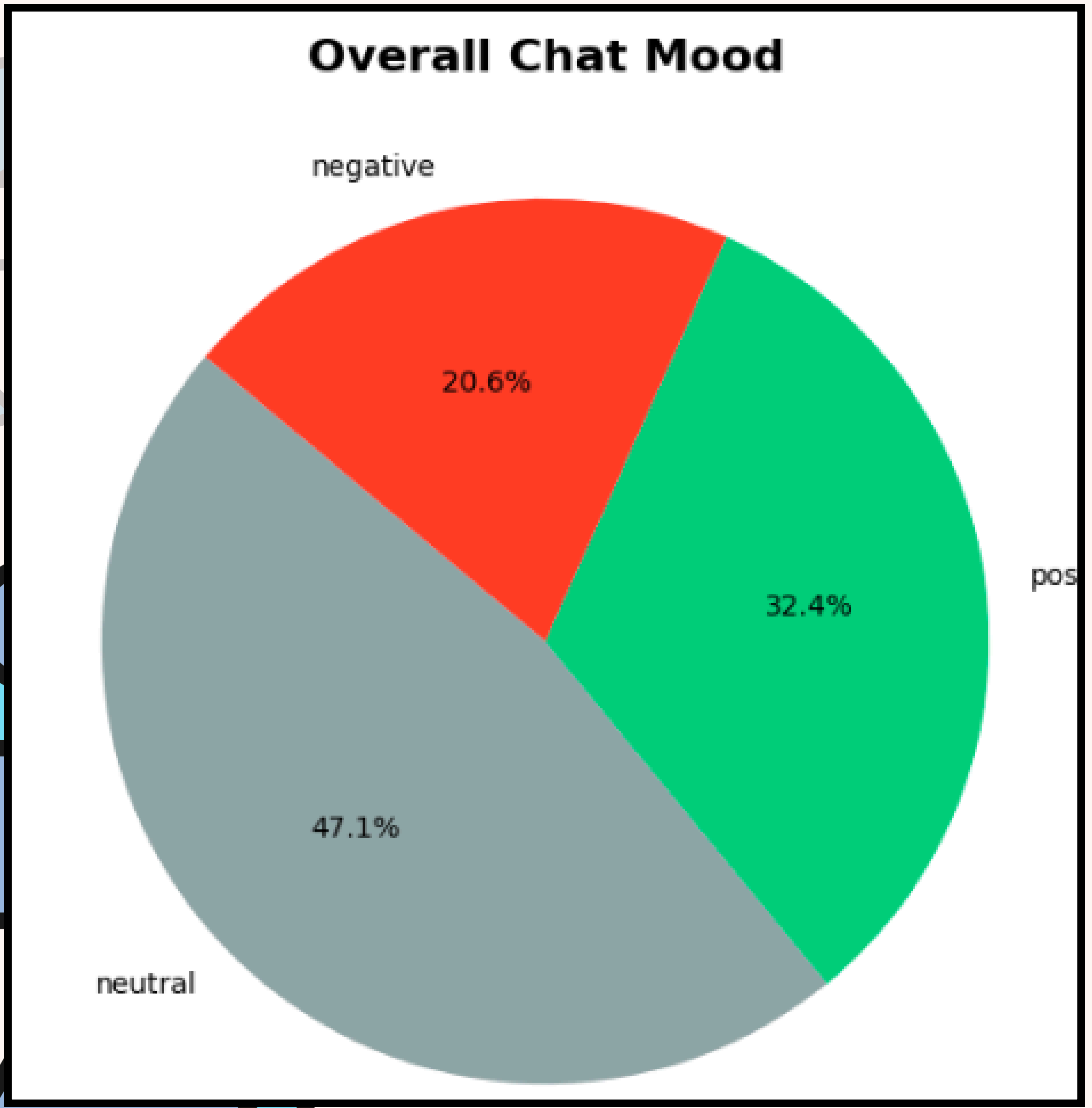
Bar graph (mood by person)



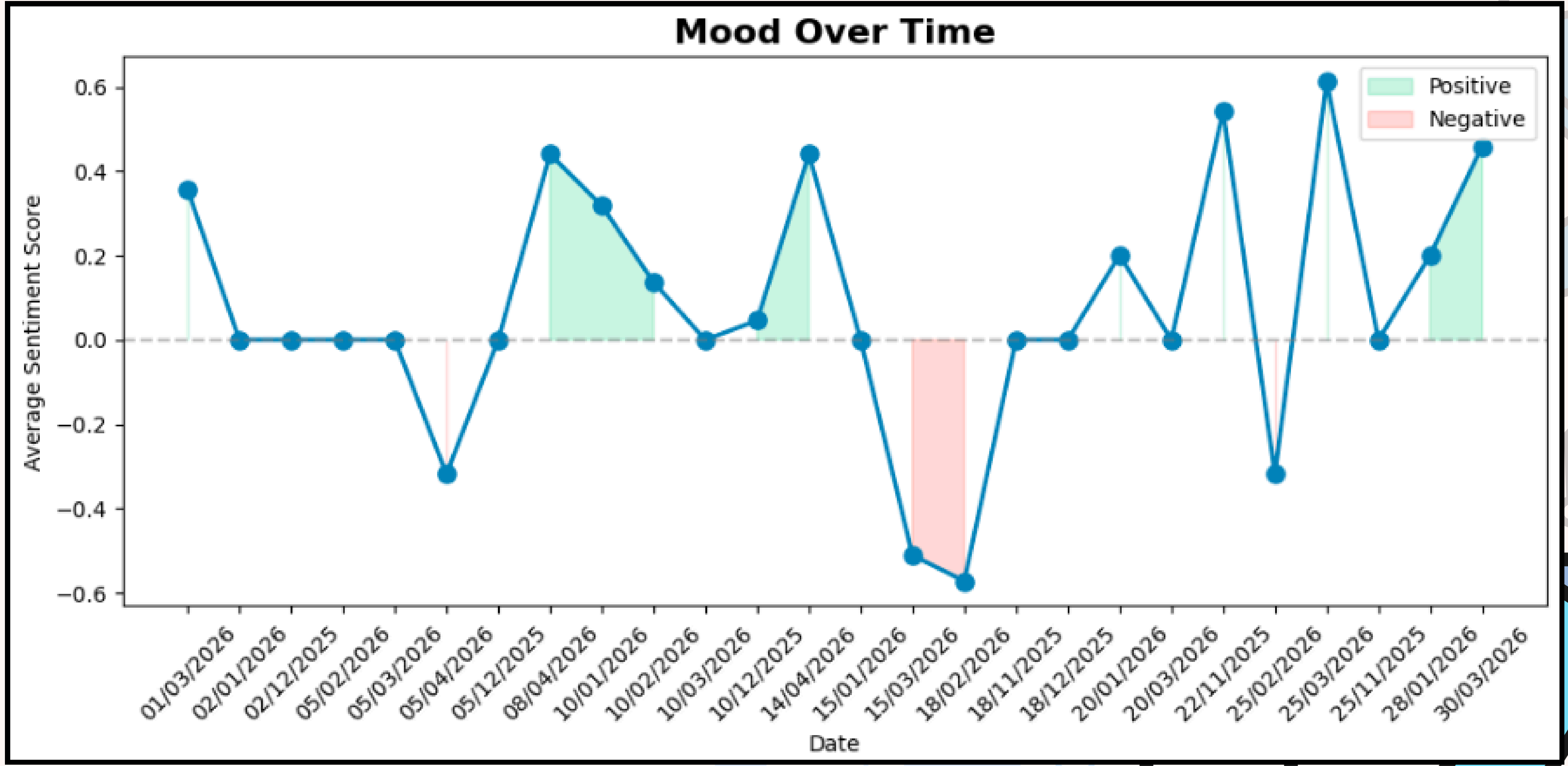
Word cloud

Visualisation 1 & 2 (chat 2)

Pie chart (overall mood)

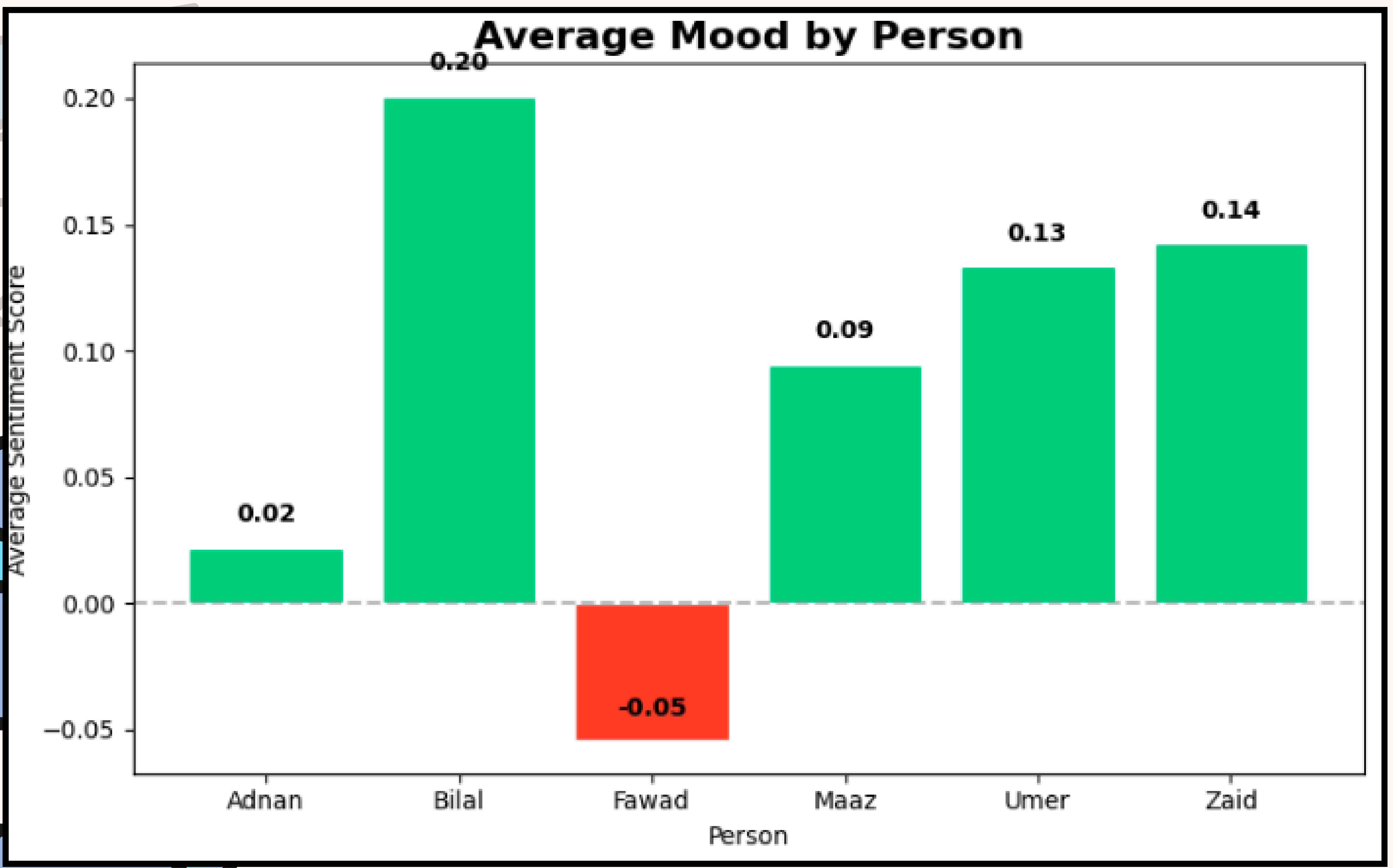


Line graph (mood over time)

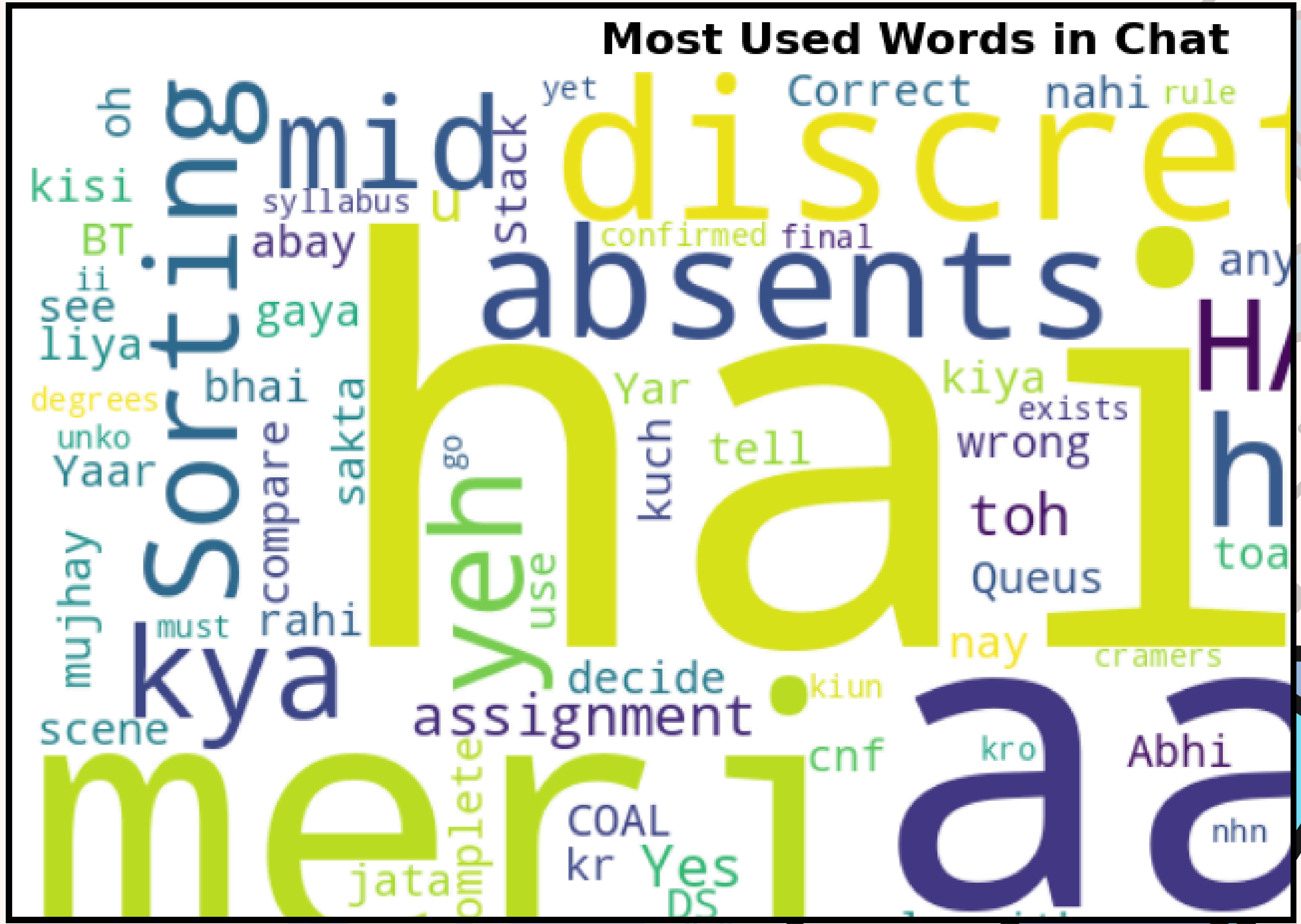


Visualisation 3 & 4 (chat 2)

Bar graph (mood by person)



Word cloud



Limitations and future improvements

Current Limitations:

- Naive Bayes trained on English
- Partial accuracy on Roman Urdu
- Very short messages often lack sufficient context to be classified reliably.

Future Improvements:

- Train on multilingual Urdu/Roman Urdu dataset
- Add fine-grained emotions (joy, anger, sadness)
- Build a web interface using Streamlit for non-technical users

Conclusion

✓ Built a complete end-to-end NLP pipeline
✓ Naive Bayes achieved 85.06% accuracy on IMDB test set
✓ Dual model approach handled both formal and informal text
✓ Tested on dummy and real WhatsApp data successfully
✓ 4 visualizations generated automatically
✓ System is practical, relatable, and immediately usable

QnA Time

Thank You